PES University – Dept. of CSE

Lab 3: Component Modelling & Architectural Pattern Selection

Problem Statement #60 | Media, Events & Community

Podcast Guest Scheduling & Outline Builder

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CLASS: CSE G SECTION

Architecture Selection

We chose Layered Architecture for the Podcast Guest Scheduling & Outline Builder System.

The system is organized into three layers: a Presentation Layer (Guest & Host Portal Component), a Business Layer (Scheduling Manager, Outline Compiler, and Notification & Calendar Service Components), and a Data Layer (Episode Database Component). Requests flow downward from the portal through the business logic to persistent storage, with each layer depending only on the layer directly below it.

Reason 1 – Clear Separation of Scheduling and Content Concerns

The system handles two fairly distinct concerns: booking logistics (time slots, blocked dates, calendar invites) and editorial content (guest bios, talking-point outlines, production sheets). A layered structure lets the Scheduling Manager and Outline Compiler evolve independently as separate business-layer components while sharing one presentation layer and one data store, without the operational overhead of running them as separate deployable services.

Reason 2 – Predictable, Low-Traffic Usage Pattern

Podcast Guest and Show Host are the only two actors, and booking/outline submissions happen in low, bursty volumes (a handful of guests per episode) rather than at internet scale. A monolithic layered system is easier to build, test, and deploy within the project timeline than a microservices split, and avoids network-latency and data-consistency issues that would add unnecessary complexity for this scale.

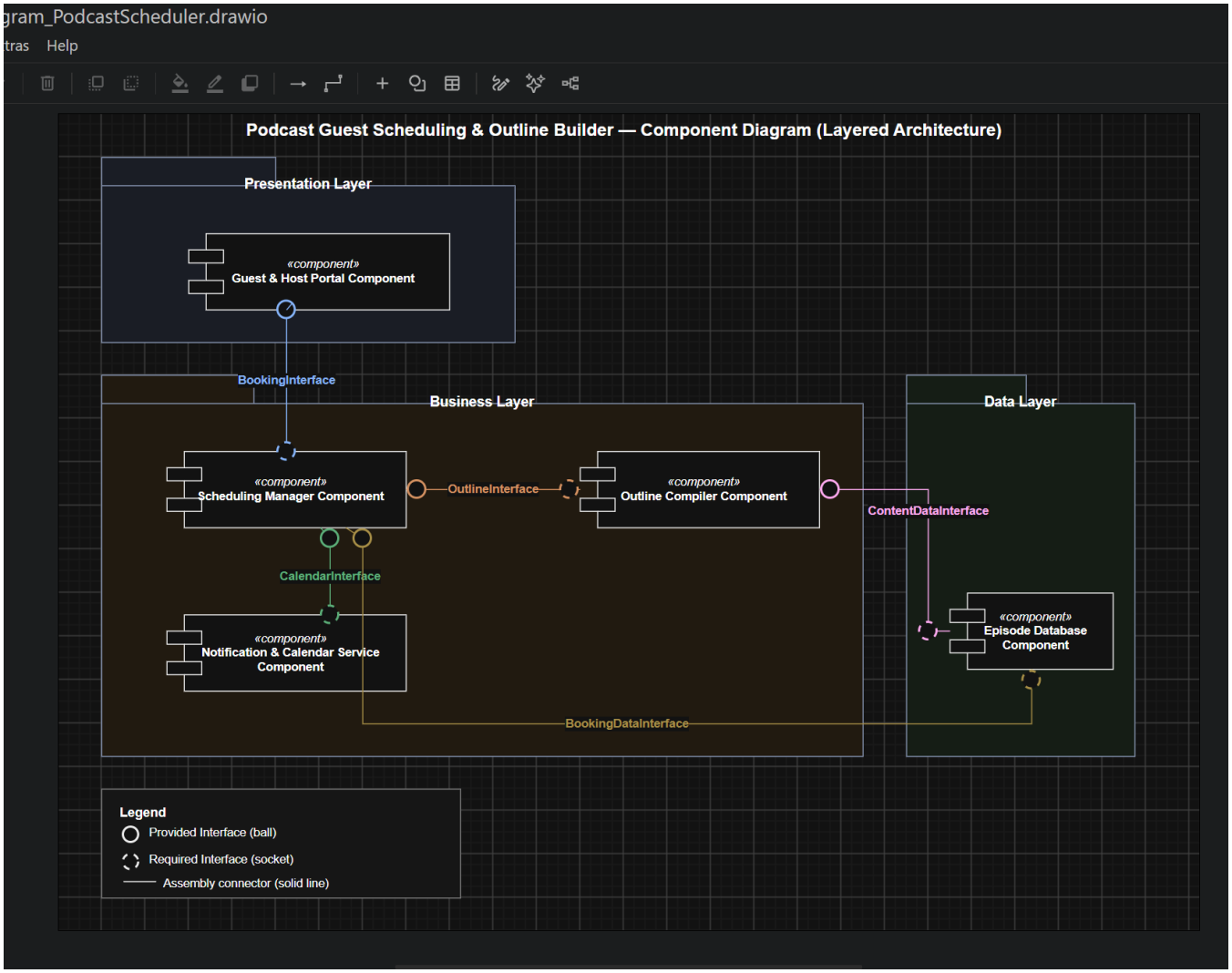
Security Advantage

Because all requests must pass through the Business Layer before reaching the Data Layer, the Scheduling Manager and Outline Compiler act as a single choke point that validates and sanitizes every booking and outline submission (e.g., rejecting bookings on blocked dates) before any data reaches the Episode Database Component. The Presentation Layer never talks to the database directly, which reduces the attack surface for injection or unauthorized data access.

Performance Benefit

Keeping the Scheduling Manager, Outline Compiler, and Notification Service within one business layer (in-process calls rather than network calls) keeps interactions such as attaching a submitted outline to the episode board and generating the timestamped production-notes PDF fast and low-latency, which directly supports the non-functional requirement that the outline generator compile a formatted production sheet in under one second.

COMPONENT DIAGRAM


gram_PodcastScheduler.drawio
tras Help
Podcast Guest Scheduling & Outline Builder — Component Diagram (Layered Architecture)
Presentation Layer
«component»
Guest & Host Portal Component
BookingInterface
Business Layer
Data Layer
«component»
Scheduling Manager Component
OutlineInterface
«component»
Outline Compiler Component
ContentDataInterface
CalendarInterface
«component»
Episode Database
Component
«component»
Notification & Calendar Service
Component
BookingDataInterface
Legend
Provided Interface (ball)
Required Interface (socket)
Assembly connector (solid line)